# ME 356
# STRENGTH OF MATERIALS II

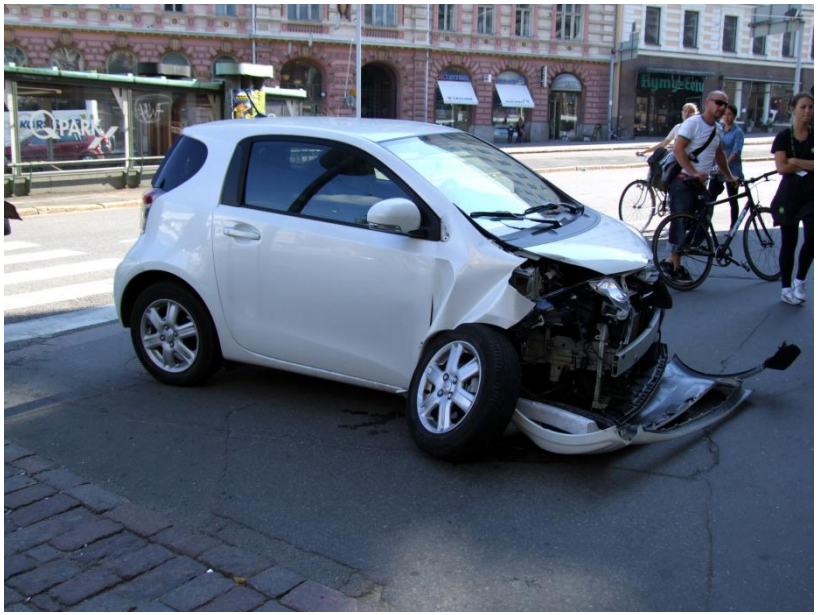


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*BSC MECHANICAL ENGINEERING, 3RD YEAR*

**DEPARTMENT OF MECHANICAL ENGINEERING**
**KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI**

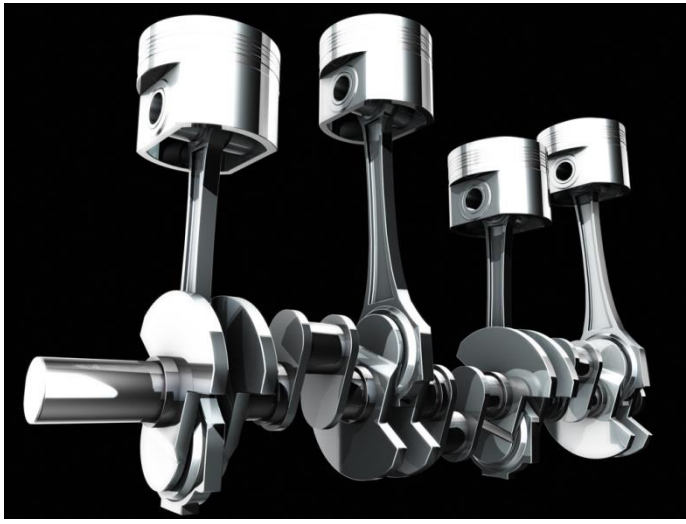
# STRENGTH OF MATERIALS II









# COURSE OUTLINE

**Topics:**

- Deflection of Beams
- Struts
- Stresses in Pressure Vessels
- Compressions, Bending and Torsion under Plastic Conditions
- Beams of Small Radius of Curvature
- Deflection of Open and Closed Helical Springs

**Reference Textbooks**

- Strength of materials 3rd Edition, G. H. Ryder
- Theory and Problems of Strength of Material, William Nash, Shaum's Outline Series
- Mechanics of Material 3rd Edition, Beer, Johnson, Jr. & DeWolf
- Mechanics of Material 3rd Edition, Gere & Timoshenko

**Grading**

- Home work 10%
- Mid Semester Exams 20%
- Final Examination: 70%

# What is a Beam

- A beam is a horizontal structural member that supports a vertical load.
- Its length is larger than its cross sectional area
- The slenderness ratio of a Beam can be 20 or more. The slenderness ratio is the ratio of the length to the least radius of gyration.
- The beam cross-section can be rectangular, circular, annular etc.
- It’s fabricated of steel, concrete, wood etc.
- It’s used in buildings, bridges, aircraft, machinery etc.

# Examples of beams

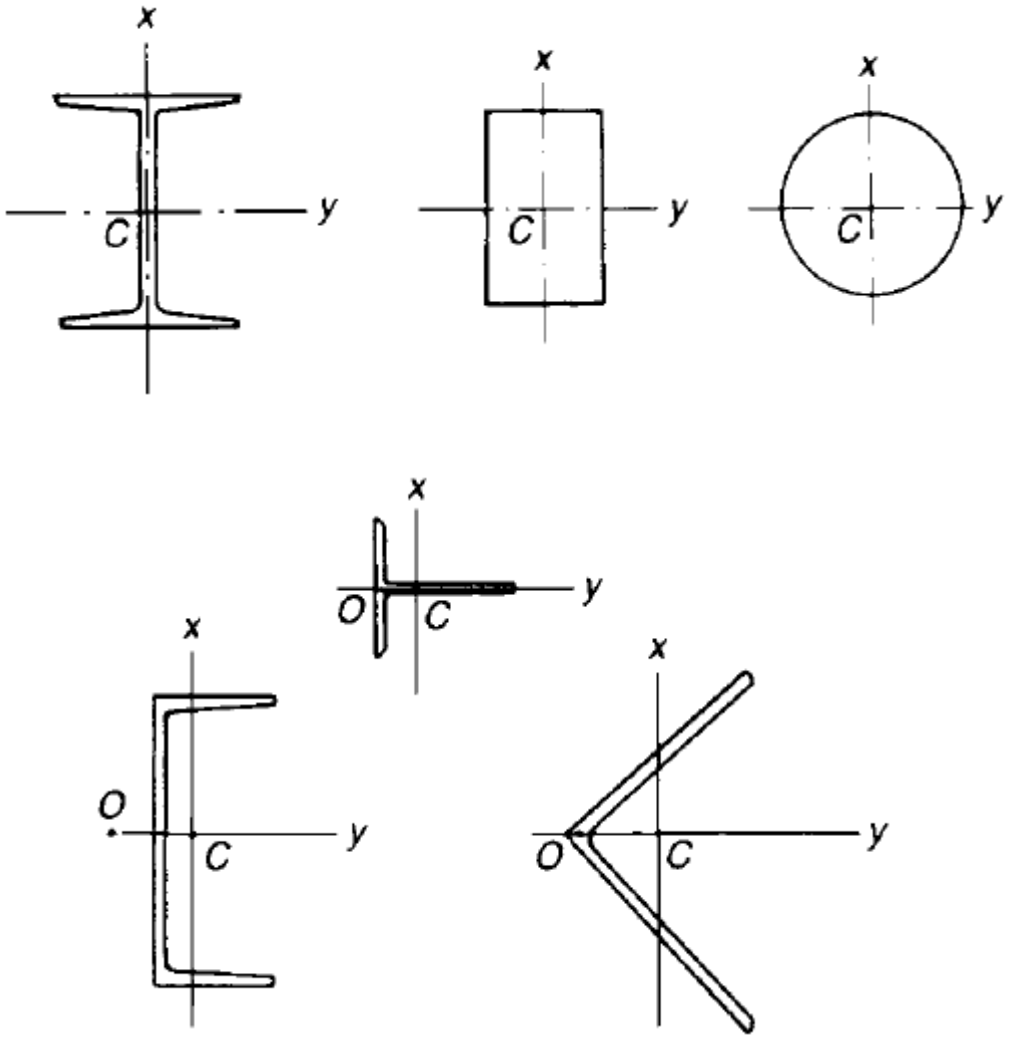


**Steel Beams**



**Bridge Beams**

# Cross-sectional areas of beams



# Deflection by Moment Integration

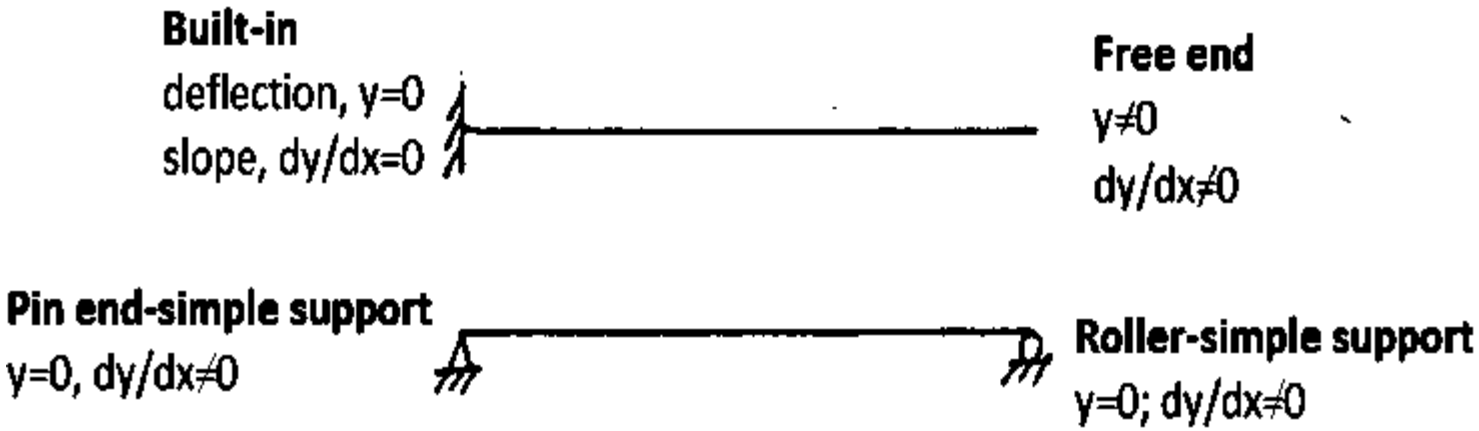
$$EI\frac{d^2y}{dx^2} = M(x);$$

$$EI\frac{dy}{dx} = \int M(x)dx + A; \text{ Slope equation}$$

$$EIy = \int\int M(x)dxdx + Ax + B; \text{ deflection equation}$$

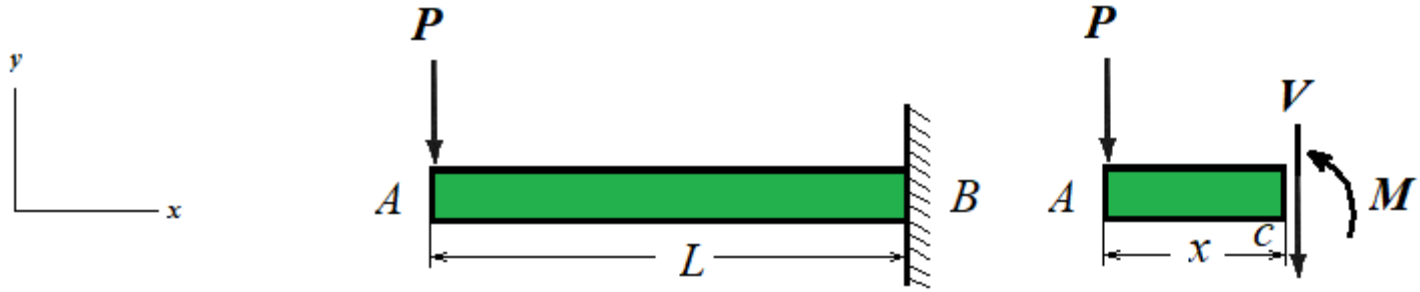
The product EI is called the Flexural Rigidity(Stiffness) of the beam.

A and B can be obtained from the boundary conditions , as follows



# Example 1-Cantilever

- Consider the cantilever shown below;




$$M(x) = -Px$$

$$EI\frac{d^2y}{dx^2} = -Px;$$

$$\frac{dy}{dx} = \frac{1}{EI}\left(\frac{-Px^2}{2} + A\right); \text{Slope}$$

$$y = \frac{1}{EI}\left(\frac{-Px^3}{6} + Ax + B\right);$$

*Boundary* Conditions

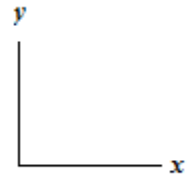
$$@\ x = l,\ \mathrm{y} = 0 \ \text{ and } \ \frac{dy}{dx} = 0; \text{ at fixed end}$$

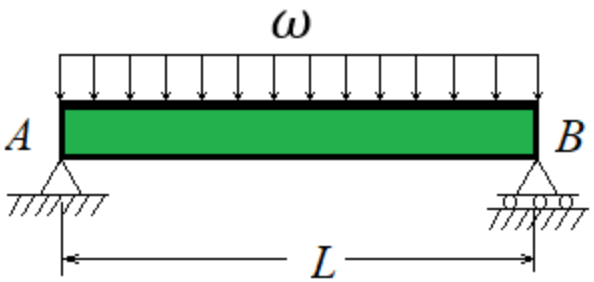
$$\therefore\ \mathrm{A} = \frac{\mathrm{Pl}^2}{2} \ \text{ and } \ \mathrm{B} = -\frac{\mathrm{Pl}^3}{3}$$

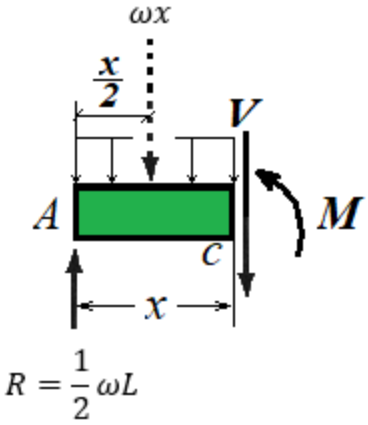
$$\frac{dy}{dx} = \frac{1}{EI}\frac{P(l^2 - x^2)}{2}; \text{Slope}$$

$$y = \frac{P}{6EI}(3l^2x - x^3 - 2l^3);$$

# Example 2- Uniformly Distributed Load on simply supported Beam.




$$M(x) = R_A x - \frac{\omega x^2}{2};\ R_A = \frac{1}{2}\omega L$$

$$EI\frac{d^2y}{dx^2} = \frac{1}{2}\omega x(L-x);$$

$$\frac{dy}{dx} = \frac{\omega}{6EI}\left(3Lx^2 - 2x^3\right) + A;$$

$$y = \frac{\omega}{12EI}\left(2Lx^3 - x^4\right) + Ax + B;$$

*Boundary* Conditions

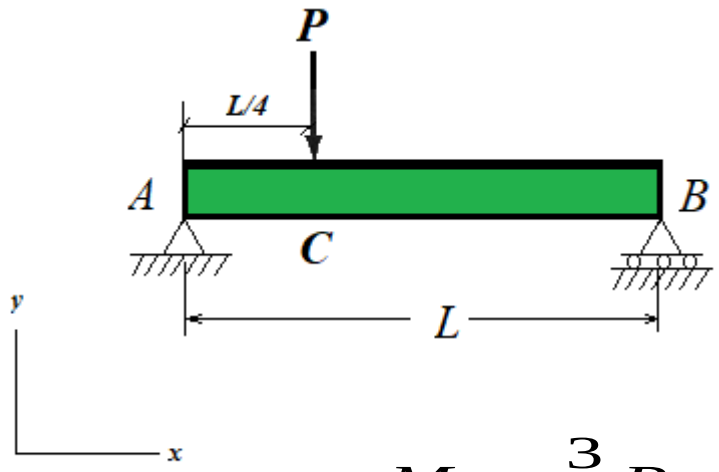
$$@\ x = 0,\ y = 0 \ \text{ and } \ @\ x = L,\ y = 0;$$

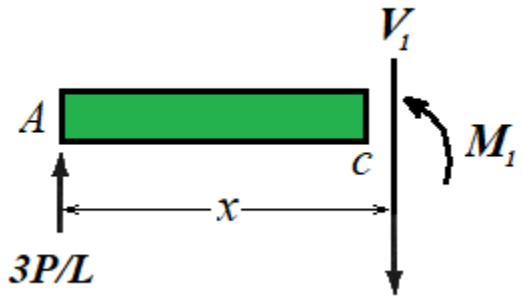
$$\therefore\ B = 0 \ \text{ and } \ A = -\frac{L^3}{12EI}$$

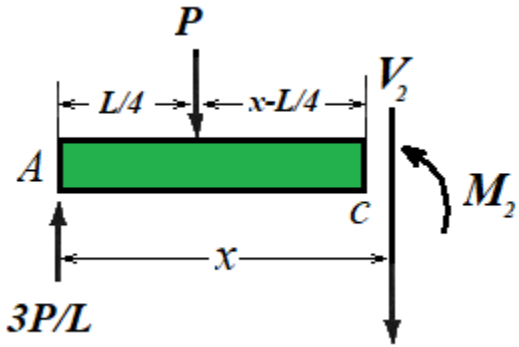
$$\frac{dy}{dx} = \frac{\omega}{12EI}(6Lx^2 - L^3 - 4x^3);$$

$$y = \frac{\omega}{12EI}(2Lx^3 - x^4 - L^3x);$$

# Example 3




$$M_1 = \frac{3}{4}Px;\ M_2 = \frac{3}{4}Px - P(x - \frac{1}{4}L);$$

$$M_2 = \frac{1}{4}P(L - x);$$

$$\frac{d^2y_1}{dx^2} = \frac{3}{4EI}Px;\ \frac{dy_1}{dx} = \frac{3}{8EI}Px^2 + A$$

$$y_1 = \frac{Px^3}{8EI} + Ax + B;$$

$$\frac{dy_2}{dx} = \frac{P}{8EI}\left(2Lx - x^2\right) + C;$$

$$y_2 = \frac{P}{24EI}(3Lx^2 - x^3) + Cx + D;$$

# Example 3 cont'd…

*Boundary* Conditions

$@\ x = 0, \mathrm{y} = 0\ ;\ \therefore \mathrm{B} = 0$

also $@\ \mathrm{x} = \frac{L}{4}, \frac{\mathrm{dy}_1}{\mathrm{dx}} = \frac{\mathrm{dy}_2}{\mathrm{dx}};$

$$\frac{3\mathrm{PL}^2}{128\mathrm{EI}} + A = \frac{7\mathrm{PL}^2}{128\mathrm{EI}} + C$$

$$\mathrm{A} - \mathrm{C} = \frac{\mathrm{PL}^2}{32\mathrm{EI}} \ldots\ldots\ldots\ldots (i)$$

$$@\ \mathrm{x} = \frac{L}{4},\ \mathrm{y}_1 = \mathrm{y}_2$$

$$\frac{3\mathrm{PL}^3}{512\mathrm{EI}} + \frac{AL}{4} = \frac{11\mathrm{PL}^3}{1536\mathrm{EI}} + \frac{CL}{4} + D$$

$$A - C = \frac{\mathrm{PL}^2}{192\mathrm{EI}} + \frac{4D}{L} \ldots\ldots\ldots\ldots (ii)$$

$$@x = L, y_2 = 0$$

$$\frac{\mathrm{PL}^3}{12\mathrm{EI}} + CL + D = 0 \ldots\ldots\ldots (iii)$$

*Substituting* $(i)$ into $(ii)$ $D = \frac{PL^3}{384EI}$

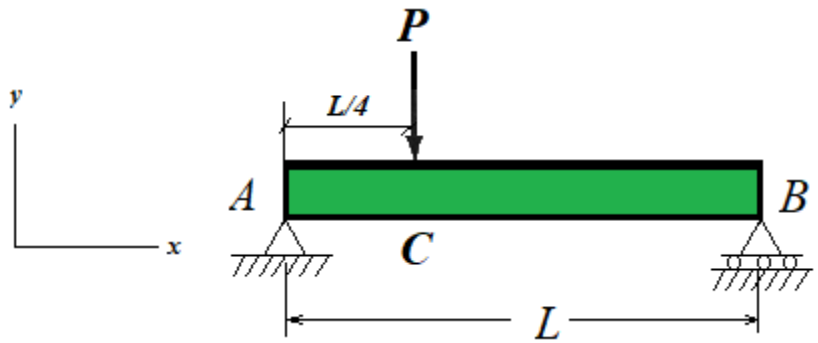
*Substituting* the value D $(iii)$ $C = -\frac{11PL^2}{128EI}$

*Substituting* the value C $(i)$ $A = -\frac{7PL^2}{128EI}$

$$y_1^l = \frac{3Px^2}{8EI} - \frac{7PL^2}{128EI} \text{ and } y_1 = \frac{Px^3}{8EI} - \frac{7PL^2x}{128EI}$$

$$y_2^l = ? \text{ and } y_2 = ?$$

# Use of Singularity Function-Macauley's Method




*It* was shown ealier that

$$M_1 = \frac{3}{4}Px;\ 0 \le x \le \frac{1}{4}L$$

$$M_2 = \frac{1}{4}P(L-x);\ \frac{1}{4}L \le x \le L$$

*These two equations may be* expressed *by a* single *equation*

$$M = \frac{3}{4}Px - P < x - \frac{L}{4} >$$

$$\therefore \frac{d^2y}{dx^2} = \frac{P}{EI}(\frac{3}{4}x - < x - \frac{L}{4} >)$$

$$\frac{dy}{dx} = \frac{P}{8EI}(3x^2 - 4 < x - \frac{L}{4} >^2) + A$$

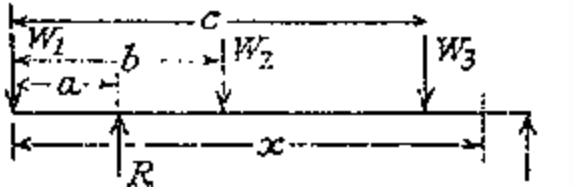
$$y = \frac{P}{8EI}(3x^3 - 4 < x - \frac{L}{4} >^3) + Ax + B$$

# Use of Singularity Function-Macauley's Method

$$@x = 0, y = 0$$

# Use of Singularity Function
# Macauley's Method

## *Concentrated loads*

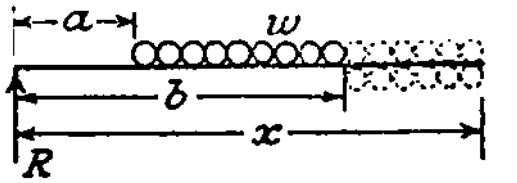



$$\therefore EI\frac{dy^2}{dx^2} = M = -W_1x + R[x-a] - W_2[x-b] - W_3[x-c]$$

$$EI\frac{dy}{dx} = \frac{1}{2}\{W_1x^2 + R[x-a]^2 - W_2[x-b]^2 - W_3[x-c]^2\} + A$$

$$EIy = \frac{1}{6}\{W_1x^3 + R[x-a]^3 - W_2[x-b]^3 - W_3[x-c]^3\} + Ax + B$$

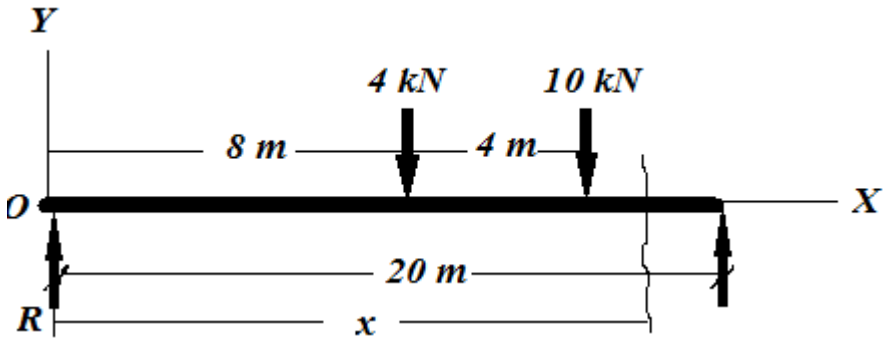
## *Uniformly distributed loads*




$$EI\frac{d^2y}{dx^2} = \left\{Rx - \frac{\omega}{2}[x-a]^2 + \frac{\omega}{2}[x-b]^2\right\}$$

# *Example 4*

A simply supported beam of span 20 m carries two concentrated loads 4 kN at 8 m and 10 kN at 12 m from one end. Calculate (a) the deflection under each load, (b) the maximum deflection.
E=200,000N/mm$^2$; I=10$^9$mm$^4$




$$\text{Re}\,action \text{ at O; R} = \frac{4\text{x}12 + 10\text{x}8}{20} = 6.4\,kN$$

$$\therefore EI\frac{dy^2}{dx^2} = M = 6.4x - 4[x-8] - 10[x-12]$$

$$EI\frac{dy}{dx} = 3.2x^2 - 2[x-8]^2 - 5[x-12]^2 + A$$

$$EIy = \frac{3.2}{3}x^3 - \frac{2}{3}[x-8]^3 - \frac{5}{3}[x-12]^3 + Ax + B$$

*Boundary* Conditions

$$@\text{x} = 0, \text{y} = 0 \Rightarrow \text{B} = 0$$

$$@\text{x} = 20\text{ m}, \text{y} = 0 \Rightarrow \text{A} = -326.5\,kNm^2$$

$(a)$ Under the 4 kN load, $\text{x} = 8\text{ m}$

$$\therefore EIy = \frac{3.2}{3}8^3 + 20\text{x} - 326.5$$

$$\text{y} = \frac{2066\text{x}10^3\,\text{x}10^9}{200\,000\text{x}10^9} = 10.3\text{ mm}$$

Similarly, under the 10 kN load, $\text{x} = 12\text{ m}$

$$EIy = \frac{3.2}{3}12^3 - \frac{2}{3}\text{x}4^3 + 12\text{x} - 326.5$$

$$\text{y} = \frac{2118\text{x}10^3\,\text{x}10^9}{200\,000\text{x}10^9} = 10.6\text{ mm}$$

# *Example 3 cont'd…*

- (b) The maximum deflection can be judged to lie between the two loads, and, omitting the term [x-12], the following equation is obtained for zero slope:

$$3.2x^2 - 2(x-8)^2 - 326.5 = 0$$

$$1.2x^2 - 32x - 454.5 = 0$$

$$x = 10.3\,\text{m}$$

(Note that this does lie within the section assumed).
Maximum deflection,

$$y_{\max} = \frac{[(3.2/2)10.3^3 - \frac{2}{3}(2.3)^3 - 326.5(10.3)]\,\text{x}10^{12}}{200\,000\,\text{x}\,10^9}$$

$$y_{\max} = 11\,\text{mm}$$

# Method 2- Strain Energy due to Bending

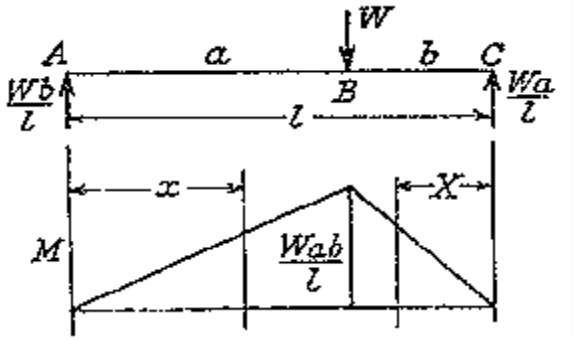
- The integration for strain energy can only be applied over a length of beam for which a continuous expression for $\boldsymbol{M}$ can be obtained.
- Also, this method is limited to cases where only one concentrated load is applied (i.e. doing work), and it only gives the deflection under this load.

# Method 2- Strain Energy due to Bending cont'd…

- For a material under-going bending within its elastic limit, the strain(elastic) energy possessed by the material is
- $U = \frac{1}{2} k \delta x^2 = \frac{1}{2} F \delta x$
- $\frac{\delta U}{\delta V} = \frac{1}{2} \frac{F}{A} \frac{\delta x}{\delta L} = \frac{1}{2} \sigma \varepsilon = \frac{1}{2E} \sigma^2$
- $U = \frac{1}{2E} \int \frac{M^2 y^2}{I} \delta V$, since $\sigma = \boldsymbol{\frac{My}{I}}$
- $U = \frac{\delta x}{2E} \int \frac{M^2 y^2}{I} \delta A = \frac{1}{2EI} \int M^2 dx,$ since $\int y^2 \delta A = I$

# *Example* 1

- *A simply supported beam of length I carries a concentrated load W at distances of a and b from the two ends. Find expressions for the total strain energy of the beam and the deflection under the load.*




Referring to the Figure above, for section AB,

$$M = (Wb/l)x$$

$$U_a = \int \left(M^2 \big/ 2EI\right) dx = \int \frac{W^2 b^2}{2l^2 EI} x^2 dx$$

$$U_a = \frac{W^2 b^2}{2l^2 EI}\left[\frac{x^3}{3}\right]_0^a = W^2 a^3 b^2 \big/ 6EIl^2$$

Similarly, by taking a variable *X* measured from C

$$U_b = \int \left(M^2 \big/ 2EI\right) dx = \int \frac{W^2 a^2}{2l^2 EI} X^2 dX$$

$$U_b = \frac{W^2 a^2}{2l^2 EI}\left[\frac{X^3}{3}\right]_0^b = W^2 a^2 b^3 \big/ 6EIl^2$$

The total strain energy is

$$U = U_a + U_b$$

$$\therefore U = \left(\frac{W^2 a^2 b^2}{6EIl^2}\right)(a+b) = \frac{W^2 a^2 b^2}{6EIl}$$

# *Example* 1 cont'd…

If δ is the deflection under the load, the strain energy must equal the work done by the load

$$\frac{1}{2}W\delta = \frac{W^2a^2b^2}{6EIl} \Rightarrow \delta = \frac{Wa^2b^2}{3EIl}$$

For a central load, *a =b =1/2,* and

$$\delta = \frac{W(1/2)^2(1/2)^2}{3EIl} = \frac{Wl^3}{48EI}$$

# *Example 2*

- *Compare the strain energy of a beam, simply supported at its ends and loaded with a uniformly distributed load, with that of the same beam centrally loaded and having the same value of maximum bending stress.*
- Solution

For a uniformly distributed load *w per unit length,* the end reactions are *wl/2,* and at a distance *x* from one end.

$$M = \left(\frac{wl}{2}\right)x - \frac{wx^2}{2}$$

$$U_1 = \int_0^l \frac{w^2x^2(l-x)^2\,dx}{4\text{x}2\text{EI}}$$

$$= \frac{w^2}{8EI}\int_0^l \left(l^2x^2 - 2lx^3 + x^4\right)dx$$

$$U_1 = \frac{w^2l^5}{240EI}$$

# *Example 2*

- For central load of *W*, (Example 1),

$$U_2 = \frac{1}{2} W\delta = \frac{W^2 l^3}{96EI}$$

$$where \quad \delta = \frac{W(l/2)^2 (l/2)^2}{3EIl} = \frac{Wl^3}{48EI}$$

Maximum bending stress=M/Z, and for a given beam depends on the maximum bending moment. Since the section modulus is the same, for the two bending stresses to be the same, then the maximum bending moment must be the same.

$$Equating \text{ maximum bending moments}$$

$$\frac{wl^2}{8} = \frac{Wl}{4} \Rightarrow wl = 2W$$

$$Ratio \; U_1 / U_2 = (8/5)$$

# ASSIGNMENT 1

(a) A cantilever has a span of 8 m and loaded with 40, 80, and 120 kN at 2, 4, and 6 m respectively from one the fixed end. The maximum stress is $700\ N\ /mm^2$ , and the beam is 300 mm deep. Find the maximum deflection and slope.

$$E = 203\ 000\ N\ /mm^2$$

(b) A beam 9 m long is symmetrically placed on two supports 6 m apart. The loading is 16.5 kN/m between the supports and 20 kN (downwards) at each end. What is the central deflection and the slope at the supports.

$$EI = 35 \text{ x } 10^{12}\ N\ mm^2$$

# Assignment 1

(c) Determine the deflections and slopes at points A, B and C located on the cantilever shown below. EI = 20 MN-m$^2$.

